

Conditional Face Attribute Generation and Classification

Final Group Project Report

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1. Introduction

This project develops a complete deep learning pipeline for attribute-controlled face generation and evaluation using the CelebA dataset, which contains 202,599 celebrity face images each annotated with 40 binary facial attributes. The system addresses two complementary problems: (1) classifying all 40 attributes from a real or generated face image, and (2) synthesizing new face images conditioned on a user-specified attribute vector.

The pipeline consists of three integrated components. First, a multi-label attribute classifier based on DenseNet169 and a Bidirectional LSTM (Irene Xia) that predicts all 40 attributes simultaneously from a 224×224 face image. Second, a Conditional Variational Autoencoder or cVAE (Zachary Cardell) that generates 128×128 RGB face images conditioned on a 40-dimensional binary attribute vector. Third, an interactive Streamlit web demo (Jiatong Peng) that integrates both models, enabling users to select attributes, generate faces, and immediately see classifier-based attribute evaluation.

The end-to-end workflow is: (1) train the classifier on real CelebA images; (2) train the cVAE on the same training split with attribute conditioning; (3) generate synthetic images and pass them through the classifier to measure attribute fidelity; (4) expose this pipeline interactively in the demo with conflict detection and real-time evaluation.

Component	Developer	Description
Multi-label Classifier (train script.py)	Irene Xia	DenseNet169 + BiLSTM for all 40 attributes; CLAHE preprocessing, disk caching, pos_weight, SWA, per-class threshold tuning
cVAE (cVAE_train_option1_lpips.py, evaluate_generated_full.py)	Zachary Cardell	128×128 conditional VAE with LPIPS perceptual loss, KL warm-up, 6-variant ablation, full 10k evaluation pipeline
Streamlit Demo (streamlit_demo.py)	Jiatong Peng	Web interface with attribute selection, hard/soft conflict detection, generation, classifier evaluation, and image upload

Table 1. Division of work across the group.

This report is organized as follows: Section 2 describes the CelebA dataset; Section 3 details both deep learning architectures; Section 4 explains the experimental setup;

Section 5 presents results from all three components; Section 6 concludes with lessons learned and future directions.

2. Description of the Dataset

2.1 CelebA Overview

The Large-scale CelebFaces Attributes (CelebA) dataset (Liu et al., 2015) contains 202,599 face images of celebrities collected from the internet. Each image is annotated with 40 binary attributes covering hair color, facial structure, makeup, accessories, and demographic characteristics. The dataset is officially partitioned into 162,770 training, 19,867 validation, and 19,962 test images, maintaining celebrity-level separation across splits to prevent data leakage.

2.2 Class Imbalance

A critical characteristic of CelebA is severe class imbalance across attributes. Positive-example rates range from 83.2% (No_Beard) to 2.1% (Bald), spanning nearly two orders of magnitude. This imbalance directly motivates the use of weighted loss functions in the classifier. Table 2 illustrates the range.

Attribute	Positive Rate (%)	Category	Challenge
No_Beard	83.2	Facial Hair	Common — easy to learn
Young	77.3	Age	Common but somewhat subjective
Smiling	48.2	Expression	Balanced
Eyeglasses	6.5	Accessories	Rare but visually distinctive
Wearing_Necktie	7.0	Accessories	Rare and often occluded
Bald	2.1	Hair	Very rare; strong visual signal

Table 2. Positive-example rates for selected CelebA attributes.

2.3 Preprocessing

Both models share a common preprocessing philosophy but differ in output resolution. For the classifier, images are processed at 224×224; for the cVAE, at 128×128. The shared steps are:

- CLAHE (Contrast Limited Adaptive Histogram Equalization) applied in LAB color space (clipLimit=2.0, tileGridSize=8×8) to enhance local contrast while

preserving color distribution — critical for color-dependent attributes such as Blond_Hair and Black_Hair.

- Gaussian blur (3×3 kernel) to reduce high-frequency noise.
- Resize and normalization using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]) for the classifier; normalization to $[-1, 1]$ for the cVAE decoder's tanh output.

For the classifier, processed tensors are saved to disk after the first pass using GPU-safe multiprocessing (spawn context, not fork), eliminating 3.25M redundant preprocessing operations across 20 epochs. For the cVAE, images are center-cropped before resizing.

3. Description of the Deep Learning Networks

3.1 Multi-Label Classifier: DenseNet169 + BiLSTM

Motivation

Multi-label classification requires independent binary predictions for each of the 40 attributes. A standard global-average-pooling head discards spatial ordering, but face images have a natural top-to-bottom structure (hair → forehead → eyes → nose → mouth → chin). We designed a CNN-RNN hybrid that treats the spatial rows of the feature map as a temporal sequence.

Architecture

The DenseNet169 backbone (Huang et al., 2017), pretrained on ImageNet, processes each 224×224 image through four dense blocks with growth rate $k=32$, producing a (B, 1664, 7, 7) feature map. Dense connectivity creates $L(L+1)/2$ connections for L layers:

$$x_l = H_l([x_0, x_1, \dots, x_{l-1}])$$

The feature map is reshaped into a sequence of 7 row vectors, each of dimension 11,648 (7 columns × 1664 channels), and processed by a two-layer Bidirectional LSTM (hidden size 512):

$$h_{final} = \text{concat}(h_{fwd_last}, h_{bwd_last}) \in \mathbb{R}^{1024}$$

A classification head applies Linear(1024→256) → BatchNorm → ReLU → Dropout(0.3) → Linear(256→40) to produce 40 independent logits.

Loss Function

BCEWithLogitsLoss with per-class positive weights addresses class imbalance:

$$pos_weight_c = (N - N_{pos_c}) / (N_{pos_c} + \epsilon)$$

For Bald (2.1% positive), $pos_weight \approx 46$, penalizing false negatives 46× more than false positives. This prevents the model from defaulting to always predicting negative for rare attributes.

3.2 Conditional VAE (cVAE)

Motivation

A Conditional VAE (Sohn et al., 2015) extends the standard VAE so that both encoder and decoder are conditioned on an auxiliary variable — here the 40-dimensional binary CelebA attribute vector c . At generation time, choosing c controls which attributes appear in the decoded image, while the latent z controls residual within-class variation (identity, lighting, pose).

Architecture

The encoder is a five-layer convolutional stack (3→64→128→256→512→512 channels, LeakyReLU activations, BatchNorm) that processes 128×128 RGB images. The final feature map is flattened, concatenated with a 128-dim attribute embedding $e(c)$, and projected to μ and $\log\sigma^2$ for a 256-dimensional latent:

$$q_{\phi}(z|x, c) = N(\mu_{\phi}(x, c), \sigma^2_{\phi}(x, c))$$

Sampling uses the reparameterization trick: $z = \mu + \sigma \odot \epsilon$, $\epsilon \sim N(0, I)$. The decoder mirrors the encoder with five transposed-convolutional blocks (512→512→256→128→64→3), concatenating z with the same $e(c)$ to reconstruct the image via tanh output. Total parameters: approximately 21 million.

Loss Function

The training objective combines three terms:

$$L = MSE(x, \hat{x}) + 1000 \cdot LPIPS(x, \hat{x}) + \beta(t) \cdot KL(q(z|x, c) \parallel N(0, I))$$

LPIPS (Learned Perceptual Image Patch Similarity) compares deep features from a pretrained AlexNet rather than raw pixels, recovering sharper reconstructions that pixel-wise MSE alone cannot produce. The weight of 1000 was chosen empirically. The KL weight $\beta(t)$ is linearly annealed from 0 at epoch 0 to 1.0 at epoch 10 to prevent posterior collapse — a failure mode where the encoder drives KL to zero and the decoder ignores z .

3.3 Streamlit Demo

The interactive demo integrates both models with three functional pages: (1) attribute selection with real-time conflict detection, (2) face generation with classifier evaluation,

and (3) image upload with direct attribute classification. The demo runs the complete pipeline — cVAE generation → preprocessing → classifier inference → per-attribute probability display — in a single user interaction.

4. Experimental Setup

4.1 Data Splits

Both models use the official CelebA train/val/test partition (162,770/19,867/19,962). The validation set is used for hyperparameter tuning and threshold selection; the test set is held out for final evaluation only. Classifier thresholds are tuned on the validation set and never on the test set to prevent data leakage.

4.2 Classifier Training Configuration

Hyperparameter	Value	Rationale
Image size	224 × 224	Standard ImageNet resolution for DenseNet169
Batch size	32	Fits 8 GB GPU; stable gradient estimates
Learning rate	1e-4	Standard fine-tuning rate; prevents catastrophic forgetting
Optimizer	AdamW (wd=1e-2)	Fixes Adam's broken weight decay
Scheduler	CosineAnnealingWarmRestarts (T_0=5)	Periodic restarts to escape local minima
Max epochs	20 (early stopping patience=8)	SWA needs epochs 8+ to accumulate averages
SWA	From epoch 8, swa_lr=5e-5	Weight averaging for flatter loss basin
Gradient clipping	max_norm=1.0	Prevents explosion from large pos_weight values
AMP	autocast + GradScaler	~2× throughput with float16 forward pass

Table 3. Classifier training hyperparameters.

4.3 cVAE Training Configuration

Hyperparameter	Value	Rationale
Image size	128 × 128	4× more pixels than 64×64; better attribute detail
Batch size	128	Larger batch stable for VAE KL estimation
Learning rate	1e-4 (Adam)	Standard VAE training rate

Latent dimension	256	Balance between capacity and regularization
KL warm-up	$0 \rightarrow 1.0$ over 10 epochs	Prevents posterior collapse in early training
LPIPS weight	1000	Dominant reconstruction signal for sharper faces
Epochs	20	Final checkpoint used (not best val loss)

Table 4. cVAE training hyperparameters.

4.4 Evaluation Metrics

The classifier is evaluated on macro-averaged F1, micro F1, and Hamming loss on the 19,962-image test set. Per-class F1 reveals which attributes are hardest to classify. For the cVAE, we generate 10,000 images conditioned on real validation-set attribute vectors, run them through the classifier with Irene’s per-attribute thresholds, and compute per-attribute F1 against the target conditioning vector. The gap (Δ) between generated-image F1 and real-image F1 isolates conditioning fidelity from raw classifier accuracy. FID measures distributional image quality.

4.5 Per-Class Threshold Tuning

For each of the 40 attributes, we sweep decision thresholds from 0.1 to 0.9 (step 0.05) on the validation set, selecting the value that maximizes per-class F1. A fixed 0.5 threshold fails for rare attributes — for Bald, the model outputs probabilities around 0.1–0.3 even for true positives, so 0.5 would classify nearly all images as negative. These tuned thresholds are then applied identically to both the classifier test evaluation and the cVAE evaluation pipeline.

5. Results

5.1 Classifier Performance on Real Images

Table 5 reports overall classifier performance on the held-out test set of 19,962 images.

Metric	Value	Interpretation
Macro F1 (test)	0.7413	Equal weight to all 40 attributes; rare attributes included
Micro F1 (test)	0.8016	Frequency-weighted; common attributes dominate
Hamming Loss (test)	0.0964	~9.6% of attribute-image pairs mispredicted; ~3.9 errors per image

Table 5. Overall classifier performance on the CelebA test set (19,962 images).

The gap between micro F1 (0.8016) and macro F1 (0.7413) reflects class imbalance: common attributes (No_Beard: F1=0.978, Young: F1=0.924) dominate the micro average, while rare subjective attributes drag down the macro. SWA contributed +0.012 val F1 improvement at no additional training cost.

5.2 Per-Attribute Classifier Results

Table 6 reports per-attribute F1, precision, and recall ranked by F1 score. Performance spans from 0.979 (Male) to 0.465 (Narrow_Eyes), a range of 0.514 points.

Rank	Attribute	F1	Precision	Recall	N Positive
1	Male	0.9788	0.9770	0.9806	7,715
2	No_Beard	0.9784	0.9725	0.9844	17,041
3	Eyeglasses	0.9779	0.9753	0.9806	1,289
4	Mouth_Slightly_Open	0.9363	0.9410	0.9317	9,883
5	Wearing_Lipstick	0.9284	0.9664	0.8933	10,418
⋮	⋮	⋮	⋮	⋮	⋮
36	Double_Chin	0.5794	0.4938	0.7010	913
37	Oval_Face	0.5654	0.4632	0.7255	5,901
38	Wearing_Necklace	0.5228	0.4885	0.5623	2,753

39	Big_Lips	0.5084	0.5941	0.4442	6,528
40	Narrow_Eyes	0.4654	0.6488	0.3629	2,968

Table 6. Per-attribute classifier results on test set (selected rows).

5.3 Key Finding: Visual Distinctiveness Over Sample Count

A critical observation is that sample count does not predict F1. Eyeglasses achieves $F1=0.978$ despite only 1,289 positive examples (6.5%) because glasses produce an unambiguous visual signal. Narrow_Eyes (2,968 positives) scores only 0.465 because the difference is subtle and annotations are inconsistent. Visual distinctiveness and label clarity matter more than sample count.

Facial hair attributes (Mustache: precision=0.48, recall=0.87; Goatee: precision=0.55, recall=0.93) systematically over-predict due to visual confusion between adjacent fine-grained categories.

5.4 Ablation: Contribution of Each Design Decision

Design Decision	Approx. Macro F1	Note
Baseline: simple 2-layer CNN	0.320	Course starting point
+ DenseNet169 pretrained	~0.589	Single largest improvement
+ pos_weight (class imbalance)	~0.604	Critical for rare attributes
+ CLAHE preprocessing	~0.618	Local contrast enhancement
+ Per-class threshold tuning	~0.627	Direct F1 optimization on val set
+ SWA (final, test set)	0.7413	Generalization boost at no extra cost

Table 7. Approximate contribution of each design decision to final classifier performance.

5.5 cVAE Generation Results

Table 8 summarizes the six cVAE variants trained across two resolutions and three loss formulations. The 128×128 MSE + $1000 \cdot \text{LPIPS}$ + KL variant was the clear winner on macro F1 and smallest Δ while remaining competitive on FID.

Variant	Res.	Loss	FID ↓	Macro F1 (gen) ↑	Avg Δ
cVAE base	64	MSE + KL (warm-up 0→0.5)	104.27	0.4095	−0.3421
σ -VAE $klw=1$	64	σ -VAE	104.27	0.4095	−0.3421
σ -VAE $klw=100$	64	σ -VAE, high KL	100.71	0.2288	−0.5228
σ -VAE $klw=100$ + LPIPS	64	σ -VAE + perceptual	97.62	0.3239	−0.4278
Option 1 + LPIPS (winner)	128	MSE + 1000·LPIPS + KL	104.41	0.4300	−0.3217

Table 8. cVAE variant comparison. FID computed against 10k real CelebA images; Macro F1 and Δ on 10k generated images evaluated by Irene’s classifier.

The winning 128×128 model achieves macro F1=0.4300 on 10,000 generated images against a real-image ceiling of 0.7517, retaining approximately 57% of the classifier’s signal. Table 9 shows the best and worst-performing attributes in generation.

Attribute	F1 (generated)	F1 (real)	Avg Δ	Category
Male	0.9630	0.9864	−0.023	Global, salient
No_Beard	0.9539	0.9770	−0.023	Global, salient
Young	0.8749	0.9188	−0.044	Global, salient
Smiling	0.8723	0.9255	−0.053	Global, salient
Mouth_Slightly_Open	0.8508	0.9379	−0.087	Global, salient
Wearing_Earrings	0.0238	0.7890	−0.765	Small-area accessory
Wearing_Necklace	0.0623	0.5315	−0.469	Small-area accessory
Rosy_Cheeks	0.0679	0.6471	−0.579	Subtle skin property
Pale_Skin	0.0850	0.6068	−0.522	Subtle skin property
Double_Chin	0.1037	0.6401	−0.536	Subtle face shape

Table 9. Best and worst attributes in cVAE generation (10k generated images).

The pattern is interpretable: global, structurally salient attributes (Male, Smiling, Young) are generated near the classifier ceiling because they change the majority of facial pixels,

aligning reconstruction loss with conditioning. Small-area accessories (Wearing_Earrings, Wearing_Necklace) and subtle skin properties (Rosy_Cheeks, Pale_Skin) collapse almost completely because their pixel footprint is too small to drive the MSE + LPIPS objective.

5.6 Streamlit Demo Results

The demo was tested across three scenarios: valid attribute combinations, hard conflicts, and soft conflicts.

For valid combinations (Smiling + Eyeglasses + Young): generated faces are visually coherent. Young achieves correct prediction (probability=0.693). Eyeglasses and Smiling are visually present but the classifier scores them lower (0.415 and 0.127 respectively), confirming the domain gap between generated and real images even when attributes are visible to humans.

For hard conflicts (Bald + Bangs): both attributes fail completely (probabilities 0.006 and 0.096, match rate=0.00). The cVAE cannot synthesize logically contradictory conditions absent from the training distribution. The conflict detection system correctly flags this combination before generation.

For soft conflicts (Female + Mustache): the model selectively represents the dominant attribute (Female, match rate=1.0) while ignoring the rare one (Mustache, probability=0.011). This demonstrates attribute priority bias toward more frequent training examples.

On uploaded real images, the classifier correctly identifies No_Beard and Young. Some inconsistencies remain (Male predicted at 0.583 for a female subject), consistent with the subjective and ambiguous nature of certain attributes.

6. Summary and Conclusions

6.1 Summary

This project developed a three-component deep learning pipeline for attribute-controlled face generation and evaluation. The DenseNet169 + BiLSTM classifier achieves macro $F1=0.7413$ on 40 attributes, a $2.3\times$ improvement over the simple CNN baseline. The 128×128 cVAE achieves macro $F1=0.4300$ on 10,000 generated images (retaining 57% of the real-image signal) with $FID=104.41$. The Streamlit demo integrates both models with hard/soft conflict detection for live user exploration.

6.2 Key Findings

- Visual distinctiveness predicts attribute $F1$ better than sample count. Eyeglasses (1,289 positives, $F1=0.978$) far outperforms Oval_Face (5,901 positives, $F1=0.565$) because glasses are visually unambiguous.
- Conditioning fidelity is highly attribute-dependent and is not captured by single-number metrics like FID . Global salient attributes (Male, Young, Smiling) are generated near the classifier ceiling; small-area accessories (earrings, necklaces) and subtle skin properties collapse almost entirely.
- The root cause of small-area failure is the reconstruction objective: $MSE + LPIPS$ losses are dominated by the majority of facial pixels, and the absence of earrings or subtle cheek color is not penalized enough to make the decoder synthesize them reliably.
- KL warm-up is essential for cVAE training. Without it, posterior collapse occurs in the first few epochs and the model ignores z entirely, producing nearly identical images regardless of the latent sample.
- SWA consistently improves classifier generalization ($+0.012$ val $F1$) at no additional training cost and is recommended as a default for multi-label classification.

6.3 Future Work

- Attribute-weighted reconstruction loss: upweight pixels in regions associated with rare accessories (ear, neck areas) to directly address the small-area failure mode.
- Auxiliary classifier-guided loss: backpropagate BCE from the classifier through the cVAE during training to close the generation-evaluation loop.
- Higher resolution (256×256): gives small accessories more pixels to land on, at the cost of substantially more compute.
- Spatial attention in the classifier: replace BiLSTM with attention over the feature map, dynamically focusing on the most relevant facial region per attribute.

- Dedicated facial hair submodel: the Mustache/Goatee/Sideburns confusion motivates a fine-grained multi-class model for this sub-category.

7. References

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8. Appendix: Code Structure

All code is available in the GitHub repository: <https://github.com/zcardell-gwu/Final-Project-Group3>

File	Developer	Description
train script.py	Irene Xia	Multi-label classifier: DenseNet169+BiLSTM, CLAHE caching, SWA, threshold tuning, full training loop
cVAE_train_option1_lpiips.py	Zachary Cardell	Winning cVAE: 128×128 architecture, MSE+LPIPS+KL loss, KL warm-up, training loop
cVAE_train_revised.py	Zachary Cardell	Revised training script with final model saving logic
evaluate_generated_full.py	Zachary Cardell	10k evaluation pipeline: generation, classifier inference, per-attribute F1, CLIP cross-check
download_data.py	Zachary Cardell	CelebA download/extract, preprocessing, attribute parsing
streamlit_demo.py	Jiatong Peng	Streamlit app: attribute selection, conflict detection, generation, evaluation, image upload

Table 10. Code files in the repository.